

CHAPTER 6

Activities & Lab Packet

Natural Resources

A complete set of labs, worksheets, writing prompts, discussions, and group projects to accompany Chapter 6. Works with both the Standard and Easy Read versions of the textbook.

 Georgia Standard S6E6

WHAT'S INSIDE

- | | | |
|----|---|------------|
| 1 | Solar Oven S'mores — Cooking with Sunshine | LAB |
| 2 | Wind Turbine Engineers — Blade Design Challenge | LAB |
| 3 | Greenhouse in a Jar | LAB |
| 4 | Renewable or Nonrenewable? | WORKSHEET |
| 5 | Conservation Solutions Match-Up | WORKSHEET |
| 6 | Reading the Temperature Trend | WORKSHEET |
| 7 | Writing Prompts (3 levels) | WRITING |
| 8 | Classroom Discussion Questions | DISCUSSION |
| 9 | Power Up Pineville! — Town Energy Plan | GROUP |
| 10 | Vocabulary Bingo | REVIEW |

1 Solar Oven S'mores — Cooking with Sunshine

LAB

OBJECTIVE

Capture renewable solar energy in a homemade oven and measure how well it works.

TIME

1 period to build + 30–60 min sunny cook time

STANDARDS

S6E6.a

Materials (per group of 3–4)

- 1 pizza box (or shoebox), aluminum foil, black construction paper, plastic wrap, tape, a wooden skewer or straw (to prop the flap)
- Thermometer; graham crackers, chocolate, marshmallows
- A sunny day (60°F+ works; full sun matters more than heat); student data sheet (next page)

Setup

Cut a flap in the box lid (leave one edge attached). Line the flap's underside with foil (reflector), the box bottom with black paper (absorber), and seal the lid opening with plastic wrap (the "greenhouse window" that traps heat — just like greenhouse gases!).

Student Instructions

1. Build your oven: foil flap, black floor, plastic-wrap window. Predict the highest temperature your oven will reach.
2. Place a s'more (cracker + chocolate + marshmallow) inside on the black paper. Record the starting temperature inside the oven.
3. Aim the foil flap so it bounces sunlight through the window onto the s'more. Prop it with the skewer.
4. Record the inside temperature every 10 minutes. Rotate the oven to follow the Sun.
5. When the chocolate is melty, record the final temperature and total time. Eat the data. Complete the sheet.

⚠ SAFETY & TEACHER NOTES

Safety: Allergy check (chocolate/graham/marshmallow); ovens can top 150°F+ inside — open carefully. Sunglasses help with foil glare.

The three jobs: foil REFLECTS light in, black paper ABSORBS it as heat, plastic wrap TRAPS the heat. Make each student point to all three.

The Ch. 4–6 bridge: the plastic wrap is literally a greenhouse-effect model — the same trapping idea behind a warming planet.

ANSWER KEY (data sheet)

1) Temperatures climb steadily, often 120–170°F inside | 2) Foil = reflects sunlight in; black paper = absorbs light as heat; plastic wrap = traps the warmed air | 3) Solar — renewable; the Sun "refills" every day and isn't used up by our cooking | 4) No fuel burned, no greenhouse gases released | Conclusion: sunlight is free, clean, and renewable, but depends on weather/time of day — a real trade-off engineers must design around.

Solar Oven S'mores — Data Sheet

Name: _____

Period: _____

My prediction: the hottest my oven will get is _____ °F

Time (min)	0	10	20	30	40	50	60
Temp inside (°F)							
What the s'more looks like							

1. What was your highest temperature? How close was your prediction?

2. Explain the job of each oven part:

Foil flap: _____

Black paper floor: _____

Plastic-wrap window: _____

3. What energy source cooked your s'more? Is it renewable or nonrenewable — and how do you know?

4. Compare your oven to a gas stove: name one advantage and one disadvantage of cooking with sunshine.

Conclusion: The plastic wrap trapped heat inside your oven. Which idea from Chapter 4 works the same way in Earth's atmosphere?

2 Wind Turbine Engineers — Blade Design Challenge

LAB

OBJECTIVE

Design, test, and improve turbine blades that turn moving air into lifting power.

TIME

50–60 minutes

STANDARDS

S6E6.a

Materials (per group of 3–4)

- 1 pencil with eraser, 1 pushpin, index cards & cardstock (blade material), scissors, tape
- String (~40 cm), a small paper cup, paperclips (the "load" to lift)
- 1 box fan (shared, teacher-controlled, steady speed)
- Student data sheet (next page)

Setup

Basic build: pin a pinwheel-style blade set to the pencil eraser; tape the string to the pencil shaft and tie the cup to the other end. When the fan spins the blades, the string winds around the pencil and lifts the cup. Hold the pencil horizontally between fingertips (or rest it across two books) so it spins freely.

Student Instructions

1. Build a standard 4-blade pinwheel and test it: how many paperclips can it lift from the floor to the pencil in 20 seconds?
2. Record the result. That's your baseline.
3. **ENGINEER TIME:** change ONE variable — number of blades, blade size, blade angle/tilt, or material. Predict what will happen, then retest with the same fan speed and 20-second limit.
4. Run at least 3 designs. Record every result, even the flops (engineers call those "data").
5. Crown your best design and explain WHY it won on the data sheet.

⚠ SAFETY & TEACHER NOTES

Safety: Fingers, hair, and strings stay clear of the fan; teacher controls the on/off and speed.

Fair test enforcement: Same fan, same distance, same time, ONE change per trial — this is the heart of the lab.

Real-world tie: Show a photo of a wind farm afterward. Real engineers run exactly this loop: design → test → measure → improve.

ANSWER KEY (data sheet)

Results vary; typically angled/tilted blades beat flat ones, and moderate blade counts beat too many (crowding blocks airflow) | 3)

Moving air (wind) → spinning blades → winding string → lifted load; real turbines spin a generator to make electricity | 4)

Renewable — the wind keeps blowing no matter how much we use | Conclusion: design choices change how much energy a turbine captures; that's why real blades are carefully shaped and angled.

Wind Turbine Engineers — Data Sheet

Name: _____

Period: _____

Trial	Design (blades: how many? size? angle? material?)	My prediction	Paperclips lifted (20 sec)
1 (baseline)	Standard 4-blade pinwheel		
2			
3			
4			

1. Which design lifted the most? What was its winning feature?

2. Why was it important to change only ONE thing per trial?

3. Trace the energy: moving air → _____ → _____ → lifted cup. In a REAL turbine, what does the spinning produce?

4. Is wind energy renewable or nonrenewable? How do you know?

Conclusion: If you had one more class period, what would you change next, and what result do you predict?

3 Greenhouse in a Jar

LAB

OBJECTIVE

Measure how a heat-trapping layer raises temperature — a model of the greenhouse effect warming Earth.

TIME

40 minutes

STANDARDS

S6E6.c

Materials (per group of 3–4)

- 2 identical thermometers
- 1 large clear jar (or zip bag / clear bowl) to cover ONE thermometer
- 1 desk lamp (or a sunny windowsill), ruler, timer
- Student data sheet (next page)

Setup

Both thermometers lie side by side in the same light, the same distance from the lamp (use the ruler). One is open to the air; one is sealed under the jar. The ONLY difference is the heat-trapping cover — that's the fair test.

Student Instructions

1. Record both starting temperatures — they should match.
2. Place the jar over thermometer B. Turn on the lamp so it shines equally on both.
3. Record both temperatures every 3 minutes for 15–18 minutes.
4. Graph both lines (open air vs. under glass).
5. Connect the model: the lamp = the Sun; the glass = greenhouse gases. Complete the data sheet.

SAFETY & TEACHER NOTES

Safety: Hot bulbs and warm glass — handle the jar by the sides when finishing.

Expected result: The covered thermometer runs several degrees warmer and keeps climbing. If the gap is small, move the lamp closer (to BOTH, equally).

The careful framing: The jar traps heat like greenhouse gases do. Then ask: "What are humans adding more of to Earth's 'jar' when we burn fossil fuels — and what should that do to the temperature line?" That's the S6E6.c argument, built from their own data.

ANSWER KEY (data sheet)

1) The covered thermometer (B) got hotter and stayed hotter | 2) The jar let light IN but trapped the heat — heat couldn't escape and mix away | 3) Lamp = Sun; jar = greenhouse gases (CO₂, methane, water vapor) | 4) More gases = a "thicker jar" = more trapped heat = rising average temperature | Conclusion: extra CO₂ from burning fossil fuels strengthens the heat-trapping layer, which matches the measured rise in global temperature over the past century.

Greenhouse in a Jar — Data Sheet

Name: _____

Period: _____

Time (min)	0	3	6	9	12	15	18
A — OPEN AIR (°F/°C)							
B — UNDER JAR (°F/°C)							

Graph both lines (time →, temperature ↑; label A and B)

1. Which thermometer ended up hotter? By how many degrees?

2. Both got the SAME light. Why did one heat up more?

3. In this model: the lamp stands for _____ and the jar stands for _____.

4. Burning fossil fuels adds extra CO₂ to the atmosphere. Using your model, what should that do to Earth's temperature — and why?

Conclusion: Scientists say global temperature has risen over the past century. Name TWO pieces of real-world evidence they use (besides thermometers under jars!).

4 Renewable or Nonrenewable?

WORKSHEET

OBJECTIVE

Classify energy resources and explain the key difference between the two groups.

TIME

15 minutes

STANDARDS

S6E6.a

Name: _____

Period: _____

Part A. Write **R** (renewable) or **N** (nonrenewable):

- | | |
|---------------------------------------|-----------------------------------|
| ____ 1. Solar energy | ____ 6. Natural gas |
| ____ 2. Coal | ____ 7. Geothermal (Earth's heat) |
| ____ 3. Wind energy | ____ 8. Nuclear (uranium) |
| ____ 4. Oil (petroleum) | ____ 9. Biomass (plant material) |
| ____ 5. Hydroelectric (flowing water) | ____ 10. Tidal energy |

Part B. Fill in the blanks:

11. The 3 fossil fuels are _____, _____, and _____.
12. Fossil fuels formed from the buried remains of ancient _____ and tiny sea creatures.
13. They take _____ of years to form — that's why they can run out.
14. "Able to be used without being used up" = _____.

Part C. Short answers:

15. What single question decides whether a resource is renewable or nonrenewable?

16. Uranium comes from the Earth and isn't a fossil fuel — so why is nuclear energy still nonrenewable?

17. Name the TWO big downsides of relying on fossil fuels.

ANSWER KEY

1) R 2) N 3) R 4) N 5) R 6) N 7) R 8) N 9) R 10) R | 11) coal, oil, natural gas 12) plants 13) millions 14) sustainable | 15) Can it refill itself faster than we use it? | 16) Uranium is a mined metal in limited supply — once used, it's gone | 17) They will eventually run out, and burning them releases greenhouse gases that warm the planet.

5 Conservation Solutions Match-Up

WORKSHEET

OBJECTIVE

Match problems facing water, soil, and air to real conservation solutions — and apply the Three R's.

TIME

15–20 minutes

STANDARDS

S6E6.b

Name: _____

Period: _____

Part A. Which resource does each solution protect? Write **WATER**, **SOIL**, or **AIR**:

- | | |
|--|---|
| _____ 1. Fixing leaky faucets | _____ 5. Rotating crops each season |
| _____ 2. Planting cover crops | _____ 6. Reducing factory emissions |
| _____ 3. Driving less, biking more | _____ 7. Taking shorter showers |
| _____ 4. Keeping chemicals out of drains | _____ 8. Planting trees on bare hillsides |

Part B. Label each action **REDUCE**, **REUSE**, or **RECYCLE**:

- | | |
|---------------------------------------|--|
| _____ 9. Carrying a refillable bottle | _____ 11. Cans melted into new metal |
| _____ 10. Old jeans become a tote bag | _____ 12. Unplugging chargers you're not using |

Part C. Short answers:

13. Define conservation in your own words.

14. Why does SOIL need protecting? (What strips it away, and why does that matter for food?)

15. Pick ONE of the Three R's and explain how it means fewer new resources get pulled from the Earth.

ANSWER KEY

1) WATER 2) SOIL 3) AIR 4) WATER 5) SOIL 6) AIR 7) WATER 8) SOIL | 9) REDUCE 10) REUSE 11) RECYCLE 12) REDUCE | 13) Using resources carefully so they last and stay clean for the future | 14) Erosion and over-farming strip fertile topsoil, and crops need that soil to grow | 15) Any reasonable answer, e.g. recycling cans means less new metal must be mined.

6 Reading the Temperature Trend

WORKSHEET

OBJECTIVE

Interpret temperature data, separate the wobble from the trend, and connect the rise to greenhouse gases.

TIME

20 minutes

STANDARDS

S6E6.c

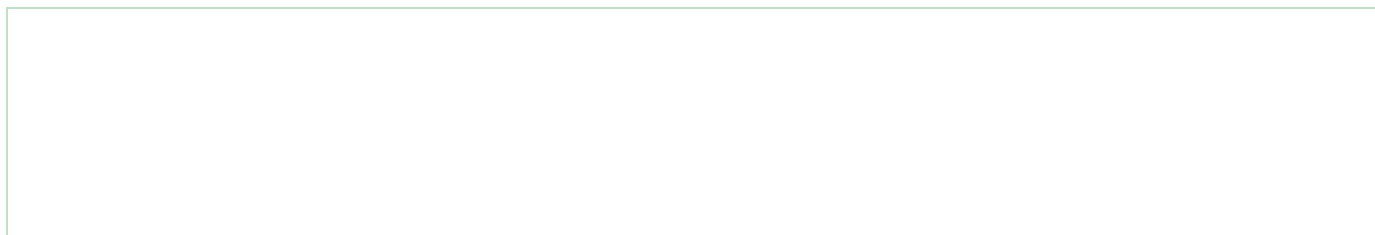
Name: _____

Period: _____

Part A. A science team shares this simplified data — Earth's average temperature compared to the long-term normal (a "+" means warmer than normal):

Decade	1920s	1940s	1960s	1980s	2000s	Today
Compared to normal	-0.2°	0.0°	-0.1°	+0.3°	+0.6°	+1.0°

1. Plot the six points on your own quick graph (or sketch the line here):



2. The line wobbles (1960s dipped!) but what does the OVERALL trend do?

3. A classmate points at the 1960s dip and says "See? No warming!" Using the words *weather/wobble* and *trend*, explain the mistake.

Part B. Connect the cause:

4. Burning fossil fuels releases the greenhouse gas _____ (chemical formula!).

5. More greenhouse gases trap more _____ near Earth's surface.

6. Besides thermometer records, name TWO other lines of evidence scientists check:

a. _____ b. _____

7. The temperature rise "lines up closely" with the rise in greenhouse gases from human activity. Why does that line-up matter to scientists making an argument?

ANSWER KEY

2) The overall trend climbs upward | 3) Single decades wobble like weather; the trend is the long-term pattern (climate) — one dip doesn't erase a century-long climb | 4) CO₂ 5) heat 6) melting glaciers; rising sea levels (also CO₂ measurements) | 7) When the suspected cause rises together with the effect across many years and many kinds of evidence, the argument that one drives the other gets much stronger.

7 Writing Prompts

WRITING

OBJECTIVE

Use scientific vocabulary to explain concepts in writing.

TIME

15–30 minutes each

STANDARDS

Cross-cutting

Level 1 — Beginner (1 paragraph)

1. **A Day Without Resources.** List five natural resources you used before lunch today (hint: water, fuels, metals...). Pick one and explain whether it's renewable or nonrenewable.
2. **Sunshine vs. Coal.** Compare solar energy and coal: where each comes from, and which one can run out. Which would you pick to power your house, and why?
3. **My Three R's Plan.** Describe one specific way YOU could reduce, one way to reuse, and one way to recycle this week.

Level 2 — Intermediate (3 paragraphs)

4. **Letter from the Future.** It's the year 2126. Write a letter back to students of today explaining which energy choices your generation is glad we made — or wishes we had made — and the science behind why.
5. **The Fossil Fuel Trade-Off.** Fossil fuels are powerful and cheap, but they run out and release greenhouse gases. Explain both sides of the trade-off, then argue what the world should do about it.
6. **Trees: The Quiet Heroes.** Explain how trees help with greenhouse gases, why cutting forests works against us twice, and propose one realistic way your community could plant or protect more trees.

Level 3 — Advanced (essay)

7. **Build the Argument.** Using evidence (temperature records, melting glaciers, rising seas, CO₂ levels), construct the scientific argument that human activity has contributed to the global temperature rise of the past century. Address one counterargument (like "it was cold last winter") and explain why it doesn't break the argument.
8. **Design the Transition.** Your country must cut fossil-fuel use in half over 30 years without leaving anyone in the dark. Write a plan: which renewables to build (and why each fits certain places), what role conservation plays, and what challenges your plan must overcome.

8 Classroom Discussion Questions

DISCUSSION

OBJECTIVE

Practice scientific thinking through verbal reasoning.

USE

Warm-ups, exit tickets, Socratic seminar

STANDARDS

Cross-cutting

TEACHER NOTES

These are designed for "think-pair-share" format. Give students 30 seconds to think, 1 minute to discuss with a partner, then call on a few groups. Some questions have no single "right" answer — push students to defend their thinking with evidence from the chapter.

Quick Warm-Ups (1–2 min each)

1. Is wood renewable or nonrenewable? (Trick question — what does the answer depend on?)
2. Name three natural resources hiding inside your phone.
3. Solar panels don't work at night. Does that make solar a bad energy source?
4. Which of the Three R's do you think saves the most resources: reduce, reuse, or recycle? Why?
5. Geothermal energy uses heat from inside the Earth. Which chapter-5 layers provide that heat?

Deeper Discussions (5–15 min each)

6. Fossil fuels are ancient sunlight stored in buried plants. Technically, will they form again? So why do we still call them nonrenewable?
7. A town must choose: a cheap coal plant now, or pricier solar and wind that pay off later. Who should make that call, and what should they weigh?
8. Glaciers store most of Earth's fresh water (Ch. 3) AND their melting is evidence of warming (Ch. 6). Connect the dots: what double problem does melting ice create?
9. "One person recycling can't fix anything, so why bother?" Build the strongest case AGAINST this claim — then the strongest case FOR it. Which wins?
10. Trees absorb CO₂, but forests are cleared for farms that feed people. How should communities balance growing food against keeping forests?

9 Power Up Pineville! — Town Energy Plan

GROUP PROJECT

OBJECTIVE

Design and defend an energy plan for a fictional town, balancing resources, geography, and cost.

TIME

2–3 class periods + presentations

STANDARDS

S6E6.a, S6E6.b, S6E6.c

Project Description

Teams of 3–4 are hired as the energy planning committee for Pineville, a town of 10,000 whose old coal plant is closing in 10 years. Each team receives the same "town fact card": Pineville has a windy ridge to the north, a river with a small dam site, lots of sunny days, nearby forests, and a tight budget. Teams must design the town's new energy mix and sell it to the "town council" (the class).

Plan Requirements

- An **energy mix** using at least 3 sources, with a pie chart showing the share of each
- A **map or poster** of Pineville showing WHERE each energy source goes — and why geography supports it
- One **conservation program** for the town (water, soil, or air) with two specific actions
- A **trade-offs box**: every plan must honestly name one weakness (e.g., "solar dips on cloudy weeks") and how the plan covers it
- A 3-minute pitch — every member speaks; the council asks one tough question per team

Presentation Day

Teams pitch; the class council scores each plan and votes to "fund" one. Debrief: what did the winning plan do that others didn't?

TEACHER NOTES

Differentiation: Provide an energy-source menu card (cost ★, reliability ★, cleanliness ★ ratings) for teams that need scaffolding; advanced teams must also estimate what happens to the town's CO₂ output.

Watch for: 100% single-source plans — push the "cloudy week / calm week" question to force a mix.

Grading rubric (suggested): Scientific accuracy (40%), required elements (25%), creativity (20%), presentation (15%).

10 Vocabulary Bingo

REVIEW GAME

OBJECTIVE

Review chapter vocabulary in a low-stress game format.

TIME

20–30 minutes

STANDARDS

S6E6 (all)

How to Play

1. Students write one word from the word bank into each square of their board, in any order. (FREE space stays free.)
2. The teacher reads a **definition** (not the word!). Students mark the square if they have the matching word.
3. First to 5 in a row calls "BINGO!" — but must read back each marked word AND its meaning to win.
4. Play again with "blackout" rules (whole board) for a longer review.

Word Bank (24 words + FREE)

natural resource	hydroelectric energy	oil (petroleum)	global temperature rise
renewable resource	geothermal energy	natural gas	reduce
nonrenewable resource	biomass	nuclear energy	reuse
sustainable	tidal energy	conservation	recycle
solar energy	fossil fuel	pollution	uranium
wind energy	coal	carbon dioxide (CO ₂)	greenhouse gas

		FREE 		

TEACHER NOTES

Read definitions straight from the Vocabulary Reference sheet (Section 6) so wording matches what students studied. Keep a checklist of called words to verify BINGOs. For struggling readers, display the word bank on the board during play.